

Computational Simulation Report

Mission OR-EXAMPLE · Dual parachute deployment · Trajectory, aerodynamics and flight-dynamics analysis

This sample report uses an OpenRocket example file (Dual_parachute_deployment.ork) paired with a placeholder motor (Icarus I / K519) purely to demonstrate the report's layout on a real dual-deployment vehicle - it is not a real Beyond UP mission and the motor pairing does not represent this airframe's actual propulsion. This mission is submitted for **LASC (Latin America Space Challenge)**. The objectives of this simulation are to predict the vehicle's ascent and descent trajectory, verify static stability throughout powered flight against the applicable safety requirements, characterize the expected landing footprint under wind uncertainty, and provide the flight cases and supporting data required for competition submission.

APOGEE AGL	MAX MACH	MIN STATIC MARGIN	FLIGHT TIME
1423.5 m	0.477	5.82 cal	59.0 s

1 General information and set-up

This sample report uses an OpenRocket example file (Dual_parachute_deployment.ork) paired with a placeholder motor (Icarus I / K519) purely to demonstrate the report's layout on a real dual-deployment vehicle - it is not a real Beyond UP mission and the motor pairing does not represent this airframe's actual propulsion. This mission is submitted for LASC (Latin America Space Challenge). The objectives of this simulation are to predict the vehicle's ascent and descent trajectory, verify static stability throughout powered flight against the applicable safety requirements, characterize the expected landing footprint under wind uncertainty, and provide the flight cases and supporting data required for competition submission.

1.1 Analysis environment

Simulation engine	RocketPy 1.13.0 (Python 3.11.15)
Application build	sample
CPU	x86_64
Operating system	Linux
Computation time	0.28 s

1.2 Model information

OpenRocket design file	Dual_parachute_deployment.ork
Motor data file (.eng)	Icarus_I_K519.eng
Drag curve source	user-supplied CSV files (power_off_drag.csv, power_on_drag.csv)

1.3 Simulation parameters

ODE solver	LSODA	Equations of motion	standard
Relative tolerance (rtol)	1e-06	Absolute tolerance (atol)	0.001 (position/velocity/quaternion components; see RocketPy's own Flight docs for the full 13-element vector)
Max. time step	unlimited (adaptive)	Min. time step	0 s
Termination condition	full flight, through ground impact or landing under recovery		

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2 Vehicle configuration and mass properties

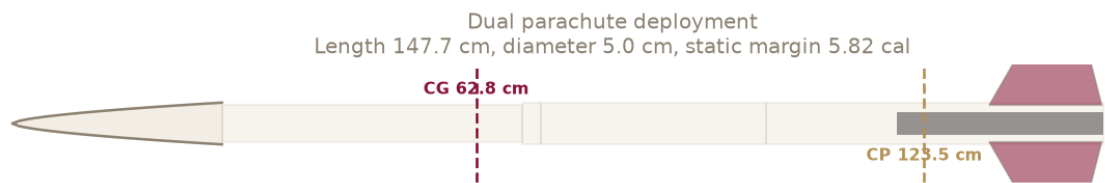


Figure 1. Vehicle side profile with center of gravity and center of pressure marked.

Figure 1 shows the vehicle's side profile with its center of gravity and center of pressure marked. Dual parachute deployment is 1.48 m long with a body diameter of 5.0 cm (reference area 0.00196 m²). The dry mass is 5.662 kg with the center of gravity at 0.628 m from the nose tip (user-entered override (LoadResult UI field)). Across the ascent the static margin ranges from 5.82 to 7.58 calibers, which is OUTSIDE the RCSM's required 1.5-4.0 cal band (FLT 4.3.5/4.3.6).

Length	147.7 cm	Diameter	5.0 cm
Reference area	0.00196 m ²	Dry mass	5.662 kg
Dry CG (from nose)	0.628 m	Mass/CG source	user-entered override (LoadResult UI field)
Static margin range (ascent)	5.82 - 7.58 cal	Stable (FLT 4.3.5/4.3.6)	NO

2.1 Parachutes

Parachute	Diameter (m)	Area (m ²)	Cd
Elliptical 12-Gore 42" Parachute	1.07	0.894	1.50
Apex 12" Drouge Parachute	0.30	0.073	0.61

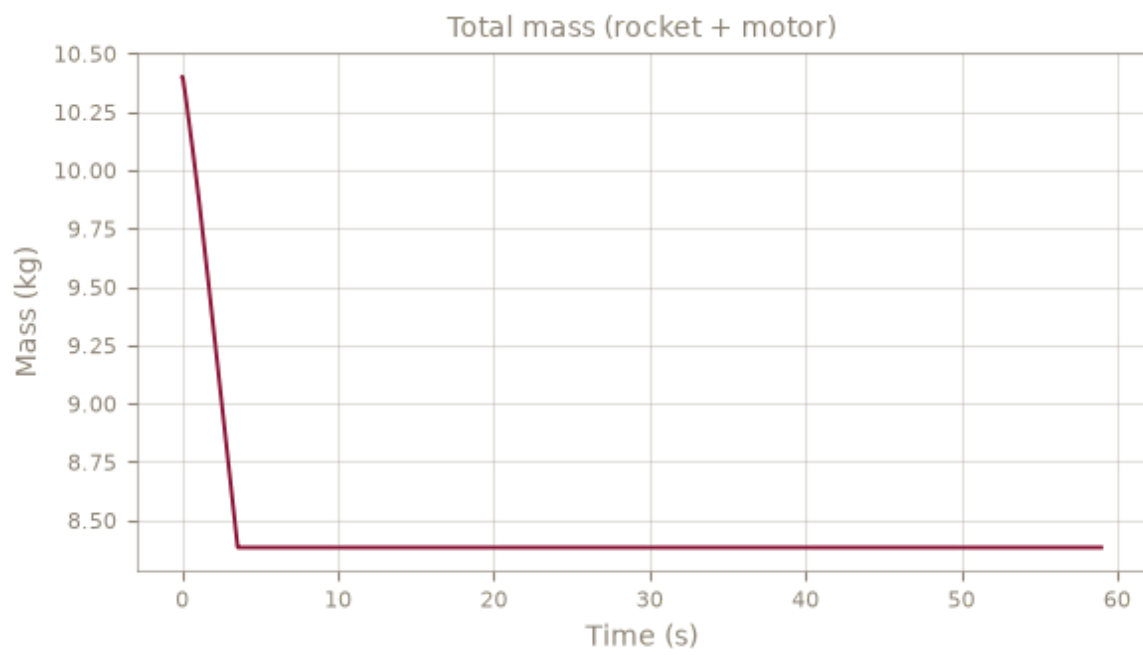


Figure 2. Total mass (rocket + motor) through the powered phase.

3 Propulsion — K519

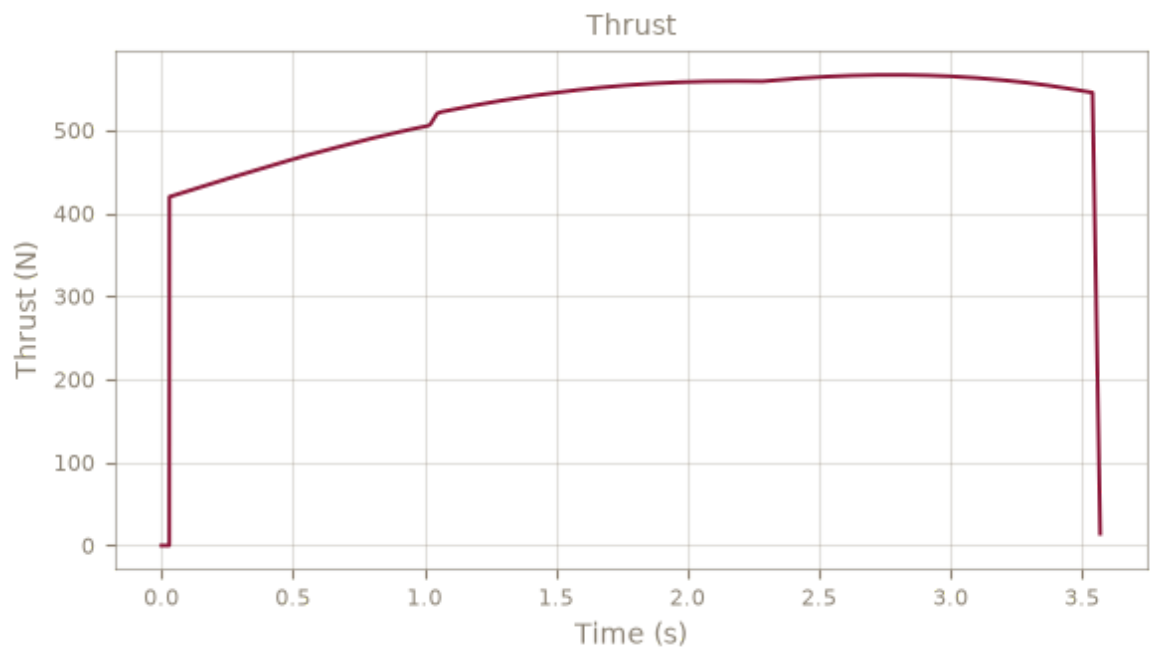


Figure 3. Thrust curve used for this simulation.

The vehicle is powered by a K519 motor (BeyondUP), delivering a total impulse of 1865.1 N*s over a 3.57 s burn (average thrust 522.4 N, peak 566.6 N). Figure 3 shows the thrust curve used for this simulation. Propellant mass is 2.017 kg out of a total loaded mass of 4.738 kg.

Motor	K519 (BeyondUP)	Total impulse	1865.1 N·s
Average thrust	522.4 N	Peak thrust	566.6 N
Burn time	3.57 s	Propellant / dry / total mass	2.017 / 2.721 / 4.738 kg

4 Trajectory — nominal and ballistic cases

Figure 4 shows the predicted altitude, velocity and acceleration through the ascent, together with the 3D trajectory and ground track. The nominal case (dual-deployment recovery) is compared against the ballistic case (no parachutes) later in Section 8, which bounds the vehicle's worst-case impact energy and downrange distance.

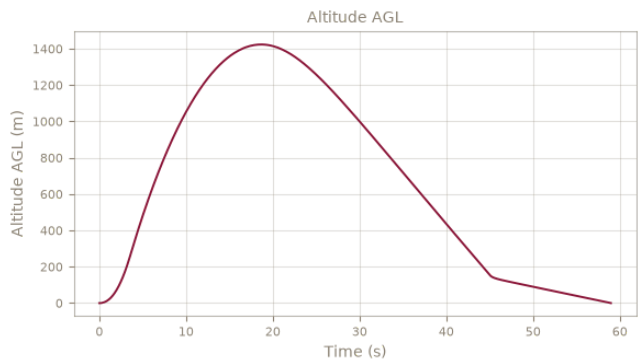


Figure 4. Altitude.

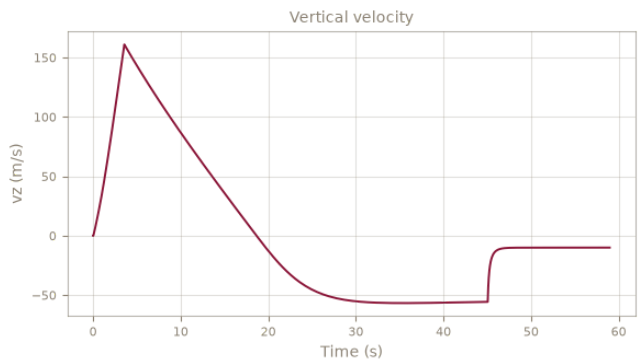


Figure 5. Vertical velocity.

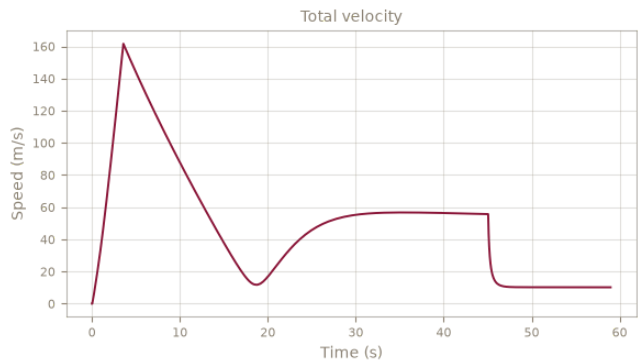


Figure 6. Total velocity.

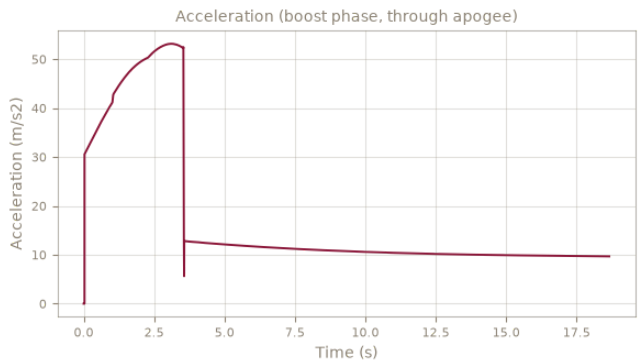


Figure 7. Acceleration (boost phase).

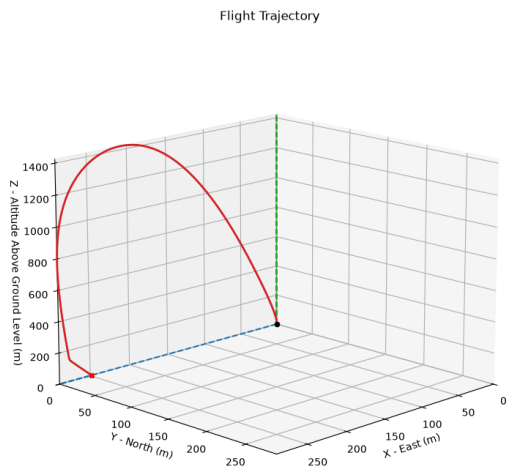


Figure 8. 3D trajectory.

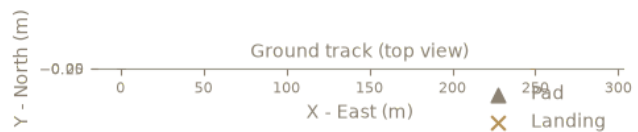


Figure 9. Ground track (top view).

5 Aerodynamics

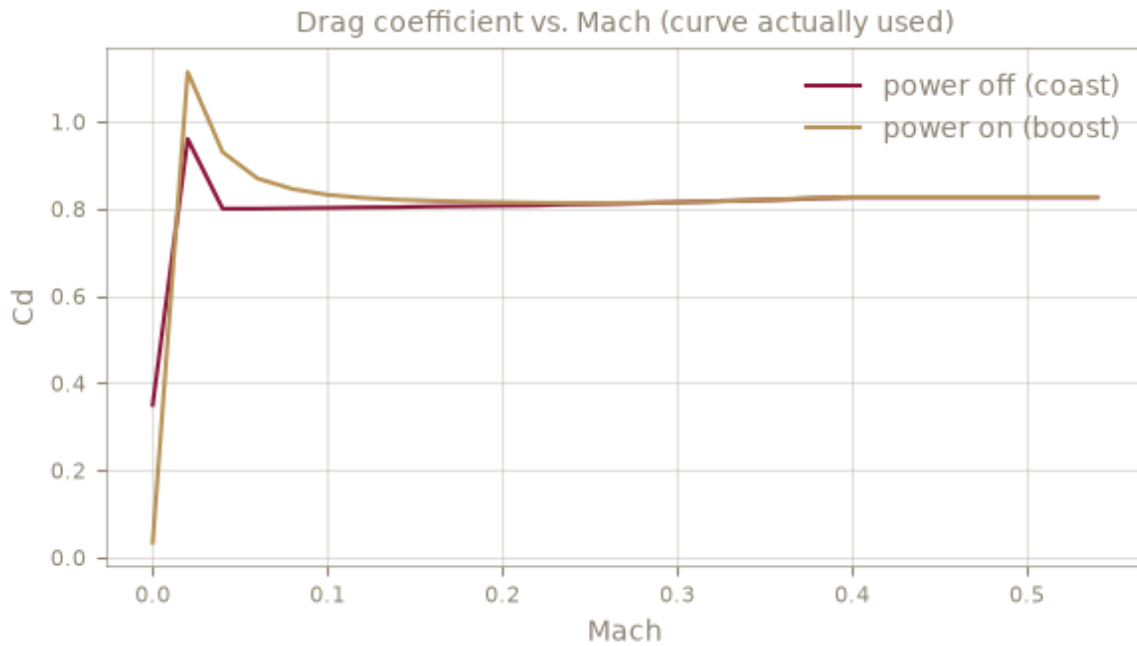


Figure 10. Drag coefficient vs. Mach number (curve actually used by the simulation).

The drag coefficient used in this simulation comes from user-supplied CSV files (power_off_drag.csv, power_on_drag.csv). Figure 10 shows the power-off (coast) and power-on (boost) drag curves actually used; the flight reaches Mach 0.48, past the drag curve's own 0.40 Mach coverage - results above that speed use a constant- C_d extrapolation, not measured drag, and should be treated with reduced confidence.

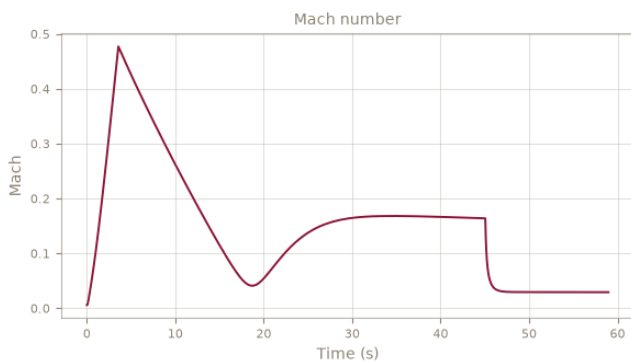


Figure 11. Mach number vs. time.

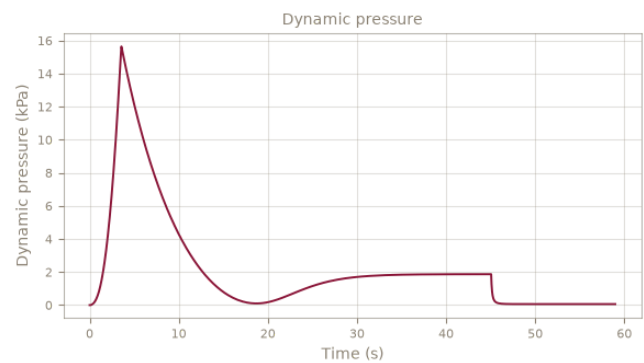


Figure 12. Dynamic pressure vs. time.

6 Stability

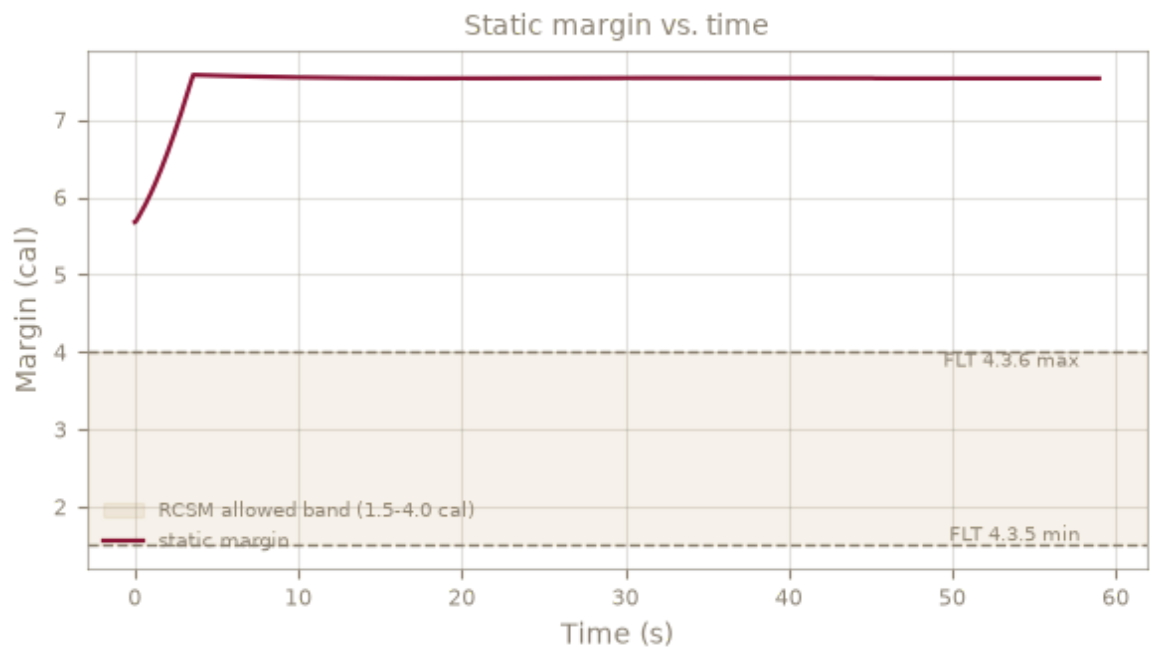


Figure 13. Static margin vs. time, with the RCSM's 1.5-4.0 cal allowed band shaded.

Figure 13 shows the static margin through the flight against the RCSM's required 1.5-4.0 cal band (FLT 4.3.5/4.3.6, shaded). The minimum margin is 5.82 cal and the maximum is 7.58 cal, so the vehicle is NOT compliant throughout the ascent. As an independent check, a hand calculation using the classical Barrowman (1966) method (Barrowman (1966), nose + fins only - body tube and transition normal-force contributions neglected (standard textbook simplification for a slender airframe).) places the center of pressure at 1.212 m from the nose tip, compared to 1.235 m from RocketPy's own (more complete) model - a difference of -1.8%.

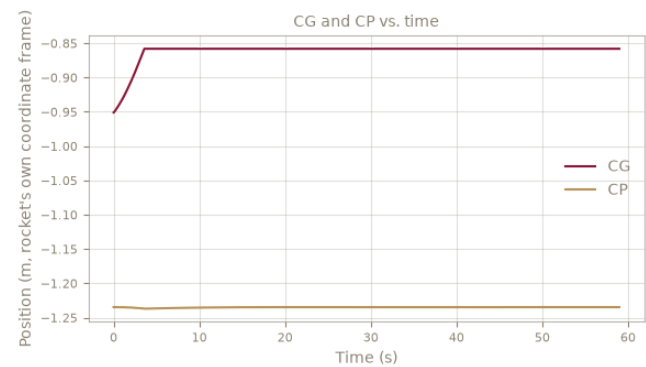


Figure 14. CG and CP vs. time.

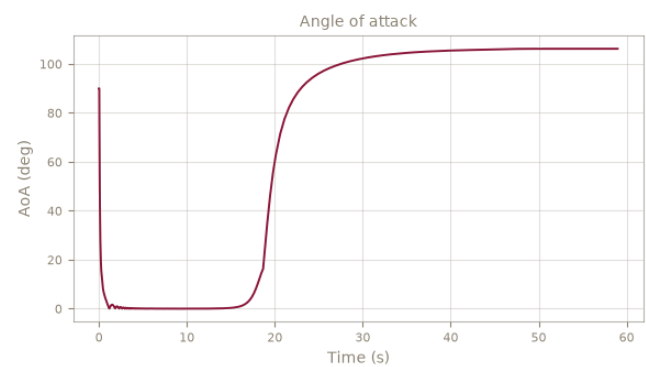


Figure 15. Angle of attack.

6.1 Independent analytical verification (Barrowman)

Barrowman (1966), nose + fins only - body tube and transition normal-force contributions neglected (standard textbook simplification for a slender airframe).

Nose $CN-\alpha$	2.00 /rad	Nose CP	0.132 m from nose
Fins $CN-\alpha$ (total, w/ interference)	13.37 /rad	Fins CP	1.374 m from nose
Total $CN-\alpha$	15.37 /rad	Hand-calculated CP	1.212 m
RocketPy's own CP (t=0)	1.235 m	Agreement	-1.8%

6.2 Comparison against OpenRocket (Simulation 1)

Static margin (ascent minimum, the KPI reported above) and stability at Mach 0.3 (t=0, OpenRocket's own default design-view snapshot) are two different quantities - see the note under the table below.

Quantity	Ours	OpenRocket	Diff
Overall length	1.477 m	n/a m	n/a
Max diameter	0.050 m	0.057 m	-12.28%
Mass without motor	5.6622 kg	1.3610 kg	+316.03%
Mass with motor (t=0)	10.4000 kg	1.6130 kg	+544.76%
CG with motor (t=0, from nose)	0.9509 m	0.9630 m	-1.25%
CP at Mach 0.3	1.2356 m	1.2010 m	+2.88%
Stability at Mach 0.3 (t=0)	5.693 cal	5.001 cal	+13.84%
Apogee AGL	1423.5 m	592.7 m	+140.17%
Max Mach	0.477	0.398	+19.86%

CP-derived rows use a 2% tolerance (RocketPy and OpenRocket use slightly different body-lift models, a documented and expected difference); every other row uses 1%.

7 Recovery and landing footprint

Stage	Diameter (m)	Area (m ²)	Cd·S (m ²)	Deploy (s)	Sim. descent (m/s)	Hand-calc descent (m/s)
Apex 12" Drouge Parachute	0.30	0.073	0.045	18.7	60.2	54.9
Elliptical 12-Gore 42" Parachute	1.07	0.894	1.341	45.1	10.2	10.0

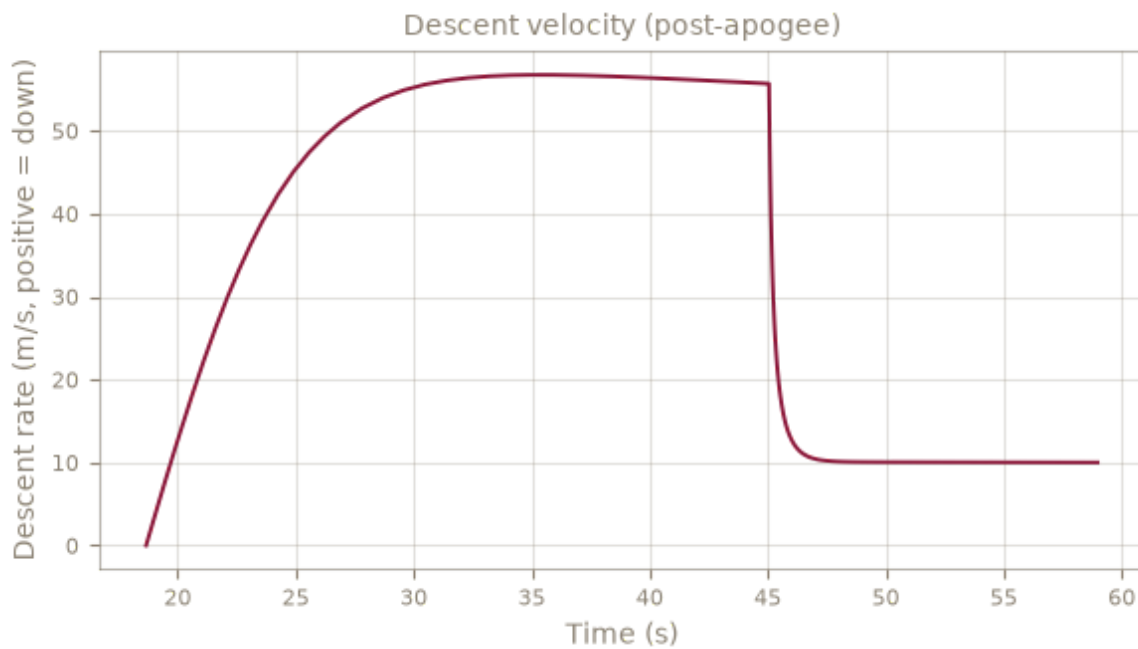


Figure 16. Descent velocity after apogee.

The recovery system uses 2 parachute(s): Apex 12" Drouge Parachute, Elliptical 12-Gore 42" Parachute. Figure 16 shows the descent rate after apogee; the simulated descent rates are compared against a hand-calculated terminal velocity for each stage in the table above, as a cross-check on the parachute Cd·S values used.

8 Flight cases (RCSM)

Case	Apogee AGL (m)	Max speed (m/s)	Rail exit (m/s)	Note
Ballistic	1423.5	161.6	13.4	
Nominal	1423.5	161.6	13.4	
DrogueOnly	1423.5	161.6	13.4	
MainAtApogee	1423.5	161.6	13.4	

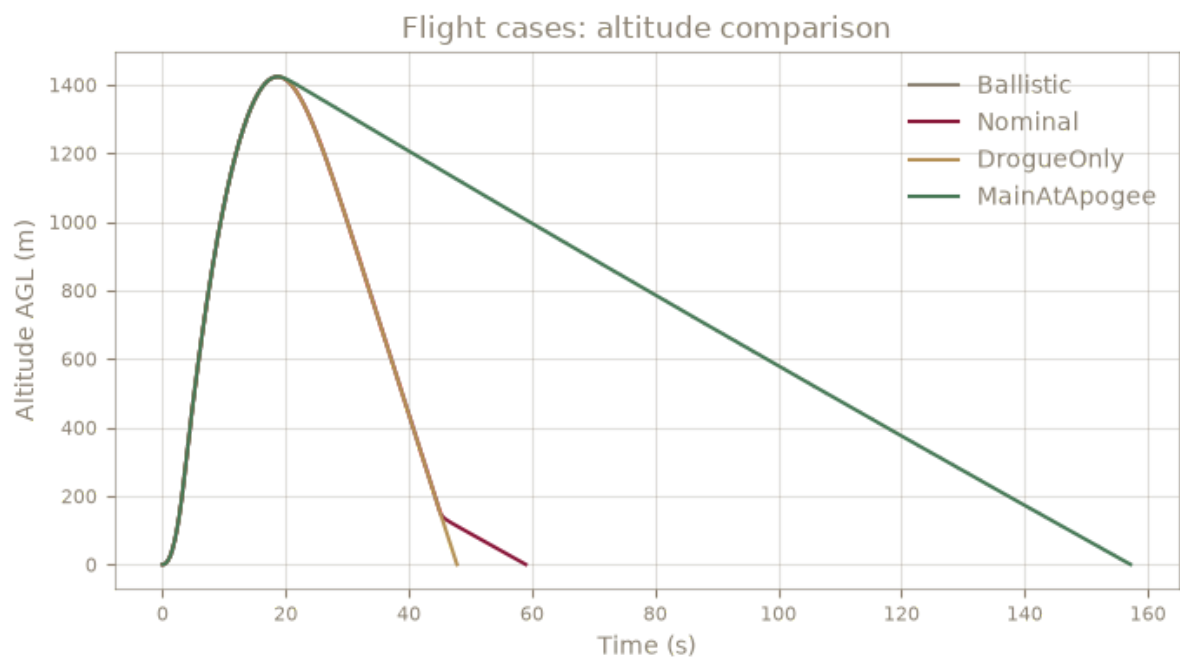


Figure 17. Altitude comparison across the required flight cases.

9 Monte Carlo dispersion

Monte Carlo dispersion analysis varies the uncertain inputs listed below across 30 simulated flights. The apogee is predicted at 1417.5 m with a 90% interval of [1390.2, 1450.0] m. The landing footprint (1/2/3-sigma ellipses) bounds where the vehicle is expected to land under wind uncertainty. With only 30 completed trajectories, this is a rough check, not a statistically meaningful dispersion estimate (fewer than 100 samples).

Low sample count. Only 30 completed trajectories - this is a rough check, not a statistically meaningful dispersion estimate.

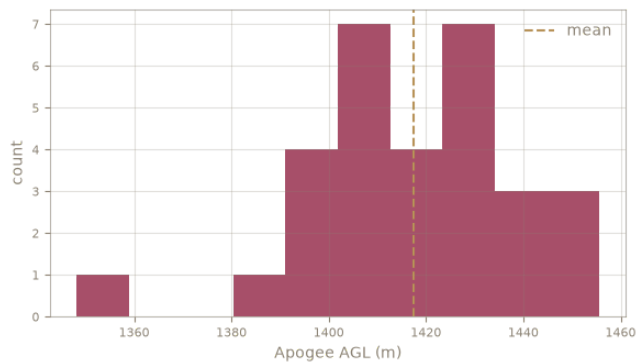


Figure 18. Apogee distribution across Monte Carlo trajectories.

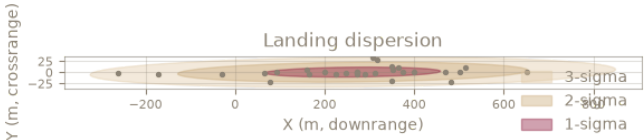


Figure 19. Landing dispersion footprint (1/2/3-sigma).

Uncertainty	Std. dev.	Source
wind_speed_ms	1	no source, low confidence - guessed as 30% of the OpenRocket-recorded wind speed
wind_direction_deg	15	no source, low confidence - typical launch-day wind direction variability
dry_mass_kg	0.1132	no source, low confidence - assumed 2% (typical scale + component-count uncertainty for a student rocket)
dry_cg_m	0.01	no source, low confidence - assumed +-1cm CG placement/measurement uncertainty. NOTE: rocketpy's StochasticRocket does NOT preserve an overridden center_of_mass_without_motor unless this is explicitly passed - found 2026-09-26, see PROGRESS.md
total_impulse_Ns	37.12	no source, low confidence - assumed 2%, typical SRAD motor-to-motor total impulse variation for KNSB
power_off_drag_factor	0.05	no source, low confidence - assumed 5% Cd curve uncertainty
power_on_drag_factor	0.05	no source, low confidence - assumed 5% Cd curve uncertainty
inclination_deg	2	no source, low confidence - typical rail-angle setup tolerance, +-2 deg
heading_deg	5	no source, low confidence - typical rail-heading setup tolerance, +-5 deg
parachute_lag_s	0.5	no source, low confidence - avionics/ejection-charge delay jitter
parachute_cd_s_factor	0.1	no source, low confidence - assumed 10% parachute Cd*S uncertainty

10 Global min-max table

Minimum and maximum of every key flight variable across the full simulated trajectory, with the time each occurs.

Variable	Minimum	at t (s)	Maximum	at t (s)
Altitude AGL (m)	-0.0	58.97	1423.5	18.66
Vertical velocity (m/s)	-56.74	35.01	161.02	3.57
Total velocity (m/s)	-0.00	0.03	161.63	3.57
Total acceleration (m/s2)	-0.00	0.03	274.81	45.05
Mach number (-)	0.006	0.02	0.477	3.57
Dynamic pressure (kPa)	0.00	0.01	15.64	3.57
Total mass (rocket + motor) (kg)	8.383	3.57	10.400	0.00
Center of gravity (m from nose)	0.858	3.57	0.951	0.00
Center of pressure (m from nose)	1.235	0.00	1.237	3.57
Static margin (cal)	5.68	0.00	7.58	3.57
Angle of attack (deg)	0.00	5.42	106.22	55.34
Thrust (N)	0.0	0.00	566.6	2.79

11 Discussion and conclusions

11.1 Discussion

The predicted apogee of 1423.5 m AGL is reached at $t = 18.7$ s, with a maximum velocity of 161.6 m/s (Mach 0.48) during the boost phase. The static margin stays within 5.82 to 7.58 calibers across the ascent, which is NOT compliant against the required 1.5-4.0 cal band.

11.2 Conclusions

Based on the results above, the vehicle is predicted to fly to 1423.5 m AGL and recover within the dispersion footprint described in Section 8. The vehicle does NOT satisfy the RCSM's static margin requirement (FLT 4.3.5/4.3.6) throughout the ascent and should be reconfigured before flight. The assumptions and data sources in this report should be reviewed before treating these numbers as final for a competition submission.

11.3 Files delivered

Item	Detail
Simulation report	report.pdf / report.docx (this document)
.ork design file	Dual_parachute_deployment.ork
.eng motor file	Icarus_I_K519.eng
Drag curve source	user-supplied CSV files (power_off_drag.csv, power_on_drag.csv)
Flight data CSV	flight_data.csv
OpenRocket-format flight data CSV	flight_data_openrocket_style.csv
Monte Carlo dispersion data	30 completed trajectories
Per-case runnable .py scripts	see the LASC/competition .zip package (CRS 10.1.3/10.1.5/10.1.6)

11.4 Assumptions and data sources

- Dry mass: 5.6622 kg (user-entered override (LoadResult UI field))
- Dry CG: 0.6279 m from nose (user-entered override (LoadResult UI field))
- Drag curve: user-supplied CSV files (power_off_drag.csv, power_on_drag.csv)
- Atmosphere: standard atmosphere model with the .ork's own recorded average wind speed/direction (not a live weather download, not a full altitude profile).
- WARNING: this flight reaches Mach 0.48, past the drag curve's own 0.40 Mach coverage - results above that speed are constant-Cd extrapolation, not measured drag.

Appendix A — Input data

A.1 Motor

Designation	K519
Manufacturer	BeyondUP
Diameter	75.0 mm
Length	280.0 mm
Delays	0
Propellant mass	2.0167 kg
Total mass (loaded)	4.7378 kg

A.2 Drag curve excerpt (evenly sampled)

Power-off (coast)		Power-on (boost)	
Mach	Cd	Mach	Cd
0.007	0.3500	0.006	0.0342
0.029	0.8220	0.020	1.1140
0.056	0.8000	0.030	1.0100
0.083	0.8010	0.040	0.9300
0.111	0.8020	0.050	0.8920
0.141	0.8040	0.061	0.8670
0.172	0.8060	0.072	0.8530
0.207	0.8070	0.084	0.8420
0.245	0.8100	0.106	0.8290
0.288	0.8140	0.130	0.8230
0.334	0.8180	0.156	0.8190
0.369	0.8220	0.212	0.8140

A.3 Parachutes

Elliptical 12-Gore 42" Parachute	1.07 m	Cd 1.50	Deploy: at 152 m AGL
Apex 12" Drouge Parachute	0.30 m	Cd 0.61	Deploy: at apogee

A.4 Atmosphere, wind and rail

Launch site	28.6100, -80.6000, 0 m MSL
Wind	2.0 m/s from 90°
Launch rail	3.0 m, 90.0° from vertical, heading 90°